Beyond Content: Surface Matching, Temporal Divergence, and the Limits of Synthetic Baselines

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Abstract

This paper tests whether broad surface matching is sufficient to reproduce the temporal structure observed in long-form Human-LLM interaction. Building on *Beyond Content: Temporal Dynamics of Long-Form Human-LLM Interaction (Baldovino, 2026),* the present study constructs a synthetic control designed to approximate the interactional envelope while deliberately excluding persistent identity, biography, long-horizon memory, explicit goals, and adaptive narrative structure. The final generator combines a frozen prompt pool with a lightweight discourse-act transition layer, introducing bounded local continuity without permitting extended adaptive arcs.

Across five independent seeds, the synthetic baseline was stable and produced assistant-side response distributions close to the empirical archive in central tendency and bulk behavior. However, synthetic user distributions remained narrower and flatter across normalized thread position than their empirical counterparts, with reduced variability and substantially lower amplification. Coupled metrics showed the same pattern: rolling amplification varied across normalized thread position in the human archive but remained comparatively flat in the synthetic condition, while post-spike recovery was more tightly regularized in synthetic runs. A lagged user-assistant coupling analysis provided secondary evidence of modestly stronger local coupling in the empirical archive.

We interpret the remaining divergence not as a simple tuning shortfall, but as a boundary of the present null family. Closing the gap further would likely require injecting higher-order temporal dependence or user-specific adaptation, which would weaken the synthetic baseline's role as a matched control. These results do not make semantic or universal claims. They show, within one empirical archive and one synthetic baseline family, that substantial surface matching alone was insufficient to reproduce the full temporal morphology of long-form Human-LLM interaction.

1. Introduction

Beyond Content: Temporal Dynamics of Long-Form Human-LLM Interaction documented large-scale temporal structure in a 25-thread Human-LLM archive. The present paper asks a narrower question: can that structure be recovered by a stateless synthetic baseline matched on broad surface interactional properties alone? This is not a semantic question. It is a structural one.

The null is straightforward. If long-form interaction morphology is fully explainable by autoregressive next-token prediction conditioned on accumulated context, then synthetic trajectories matched on broad surface statistics should exhibit comparable regime-level behavior. If they do not, then the null is, at minimum incomplete for this archive and this baseline family.

This paper does not claim model learning, agency, consciousness, or cross-session weight update. It evaluates whether a matched stateless synthetic control can reproduce the observed temporal morphology of one empirical Human-LLM archive. The synthetic generator was intentionally not optimized toward user imitation. Pushing the user-side synthetic closer to the empirical archive would risk converting the control into a proxy for the focal corpus, undermining its interpretive value as a null baseline.

2. Research Question and Hypotheses

Research question. Can a stateless synthetic baseline, matched on broad surface properties, reproduce the temporal morphology observed in the empirical long-form archive?

H0. Regime formation and morphology in long-form threads are fully accounted for by static autoregressive next-token prediction conditioned on accumulated context. Synthetic trajectories matched on broad surface statistics will reproduce indistinguishable regime-level metrics from human-coupled trajectories.

H1. Human-coupled interaction modulates regime morphology such that matched synthetic baselines remain distinguishable on one or more structural metrics, including user-side variability, amplification behavior, and recovery dynamics, without implying model learning or agency.

3. Methods

Empirical archive. The empirical condition is drawn from the long-form archive described in *Beyond Content: Temporal Dynamics of Long-Form Human-LLM Interaction* and serves as the observational anchor for the present comparison. To reduce model-family confounding, the five-seed synthetic baseline was generated with GPT-4o-mini, matching the empirical dataset at the model family level.

Synthetic baseline. The final synthetic generator combines a frozen prompt pool, anti-degeneracy guards, lognormal turn-length sampling calibrated to the sustained long-form regime, and a lightweight discourse-act transition layer. This design introduces bounded local continuity without permitting extended adaptive arcs. It preserves broad surface constraints while deliberately excluding persistent identity, biography, explicit goals, long-horizon memory, and user-specific adaptation.

Context. The synthetic condition was not context-free. It preserved autoregressive generation over a rolling conversation window and included bounded local follow-up structure. This design allowed the baseline to condition on recent interaction history without incorporating longer-horizon adaptive structure.

Metrics. Primary metrics included user length, assistant length, amplification ratio, rolling mean and rolling standard deviation, rolling amplification ratio, normalized trajectories, segment-level comparisons, post-spike recovery, and lagged user-assistant length coupling. Lagged coupling is treated as descriptive rather than causal.

Table 1. Summary of human vs. synthetic structural patterns

Metric	Human archive	Synthetic baseline (mean across 5 seeds)	Interpretive read
User length mean / SD	23.24 / 32.98	21.76 / ~4.9	Synthetic user prompts remain markedly narrower and less variable.
Assistant length mean / SD	211.89 / 118.65	210.22 / ~99.2	Assistant-side central tendency is closely matched.
Amplification mean / SD	23.38 / 42.22	~10.12 / ~5.3	Human amplification is broader and much heavier-tailed.
Rolling amplification across thirds	20.74 -> 26.65 -> 22.75	9.88 -> 10.41 -> 10.08	Human coupling changes over thread position; synthetic stays comparatively flat.
Lagged length coupling	lag-0 0.16; lag+1 0.07	lag-0 0.02; lag+1 -0.01	Supportive but secondary evidence of stronger local coupling in the archive.
Post-spike recovery	Broader excursion; slower return	Tighter excursion; faster return	Synthetic runs appear more tightly regularized after spikes.

4. Results

4.1 Assistant-side bulk behavior was closely matched

Across five independent seeds, the synthetic baseline was stable and clustered tightly on assistant-side central tendency. Synthetic assistant means ranged from 204.71 to 217.69 with a mean-of-means of 210.22, close to the empirical archive mean of 211.89. This indicates that the baseline successfully occupied the broad assistant response envelope targeted by calibration.

That match is partial rather than total. The synthetic assistant remains somewhat narrower in spread than the empirical archive. The central tendency and bulk behavior are sufficiently close that assistant-side mismatch is an incomplete explanation for the remaining divergence.

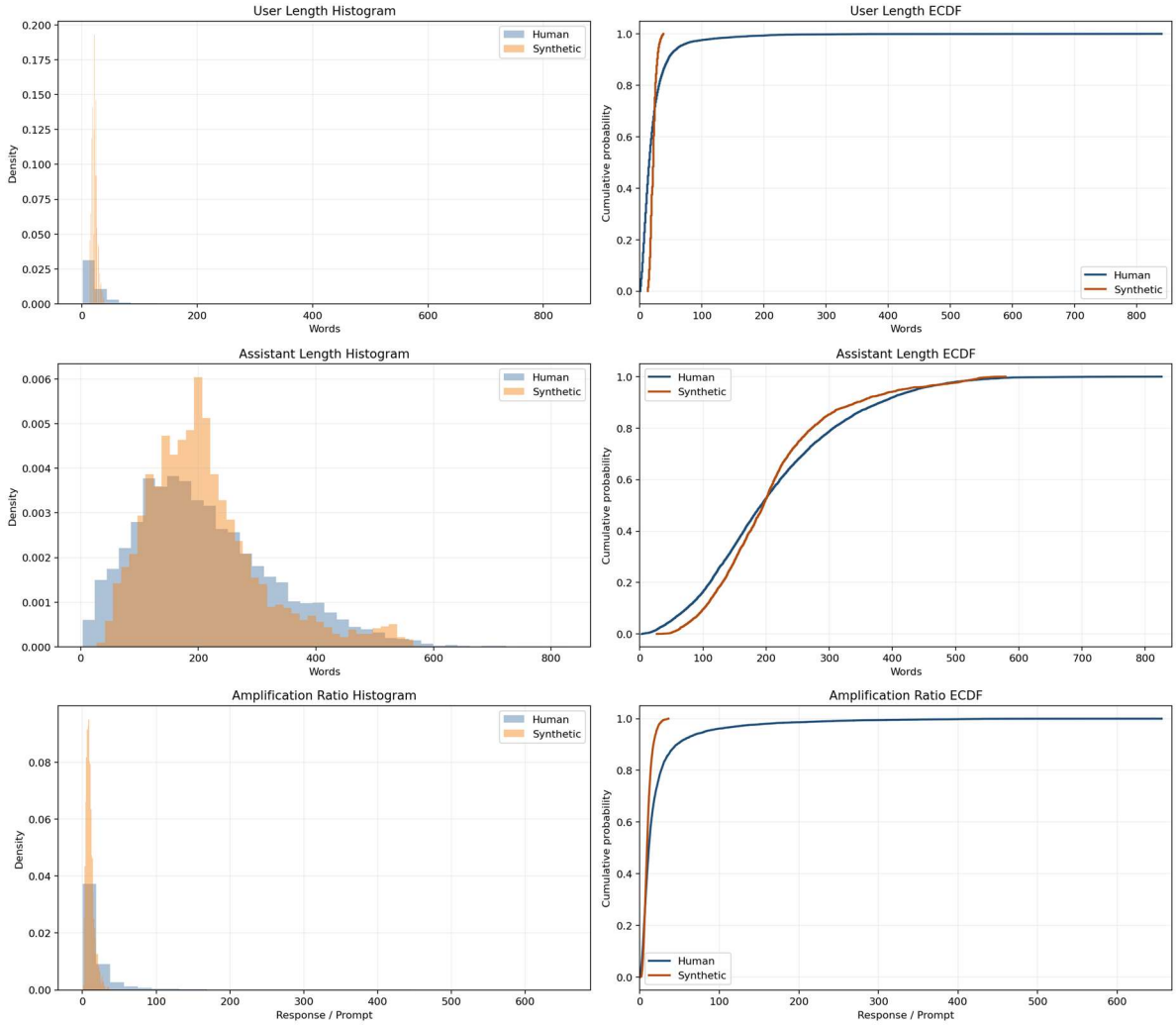

Figure 1. Distribution overlays for user length, assistant length, and amplification. The synthetic baseline approaches the assistant envelope but remains compressed on user-side variability and amplification.

4.2 User-side synthetic behavior remained compressed

The main residual mismatch appears on the synthetic user side. Human user turns in the empirical archive have a mean length of 23.24 and a standard deviation of 32.98, whereas synthetic user turns cluster around means near 21 to 22 with standard deviations near 4.6 to 5.0. The synthetic users are therefore not simply shorter; they are substantially more regular.

This regularity is visible across normalized conversation position as well. Synthetic user trajectories remain flatter than their empirical counterparts, indicating that the user-side control captures a broad surface envelope without reproducing the uneven positional structure of the archive.

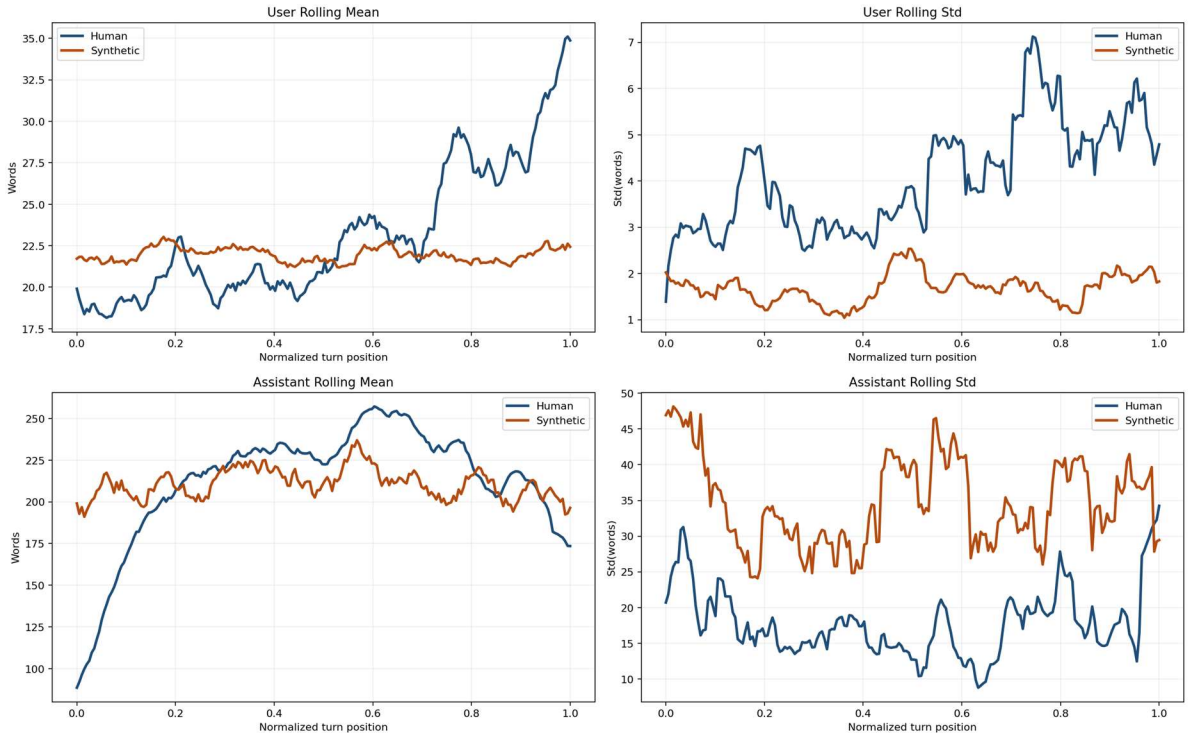

Figure 2. Rolling mean and rolling standard deviation across normalized turn position. Human threads show broader positional variation than the stateless synthetic baseline.

4.3 Coupled metrics remained divergent after surface matching

The clearest coupled residual is amplification. In the empirical archive, the response-to-prompt ratio has a mean of 23.38 and a standard deviation of 42.22, with a heavy upper tail. In the synthetic runs, amplification clusters around a mean near 10 with a standard deviation around 5.3. Surface matching on the assistant side therefore did not recover the broader user-assistant coupling structure.

Rolling amplification makes the same point in a more explicitly interactional form. In the empirical archive, rolling amplification rises across the early and middle thirds before easing later. In the synthetic condition, rolling amplification remains comparatively flat across thirds. The user-assistant relationship is therefore more positionally structured in the empirical archive than in the synthetic null.

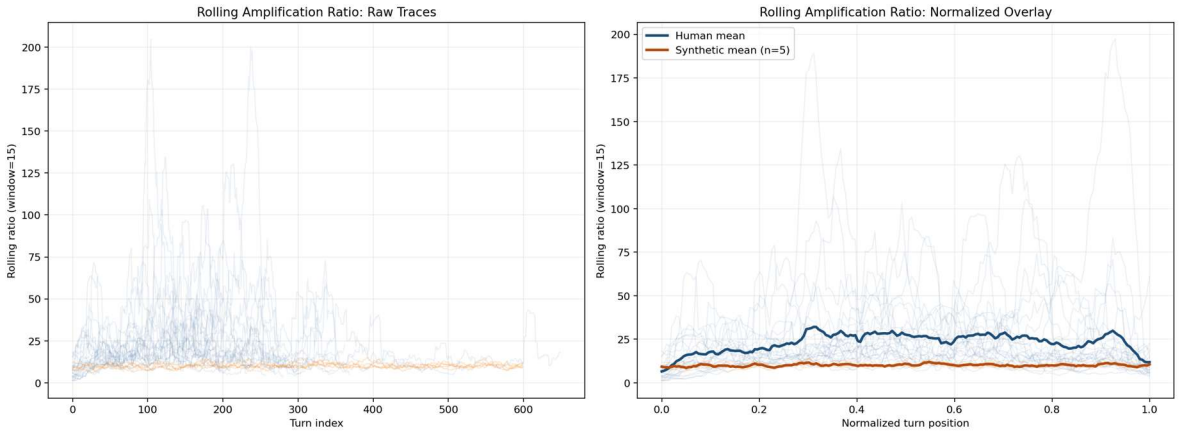

Figure 3. Rolling amplification ratio. Human rolling amplification changes across normalized thread position, while synthetic runs remain comparatively flat.

4.4 Recovery dynamics and lagged coupling provided secondary support

Post-spike recovery analysis examines how quickly rolling amplification returns toward local baseline after a detected spike. Both conditions move back toward range, but the synthetic average returns sooner and with less excursion. This is consistent with a more tightly regularized null regime rather than with runaway escalation or instability.

A lagged user-assistant length coupling analysis yielded modestly higher same-turn and short-lag association in the empirical archive than in the synthetic runs. Mean lag-0 association was 0.16 in the archive versus 0.02 in the synthetic runs, and mean lag+1 association was 0.07 versus -0.01. These differences are supportive rather than decisive and are best interpreted as convergent secondary evidence.

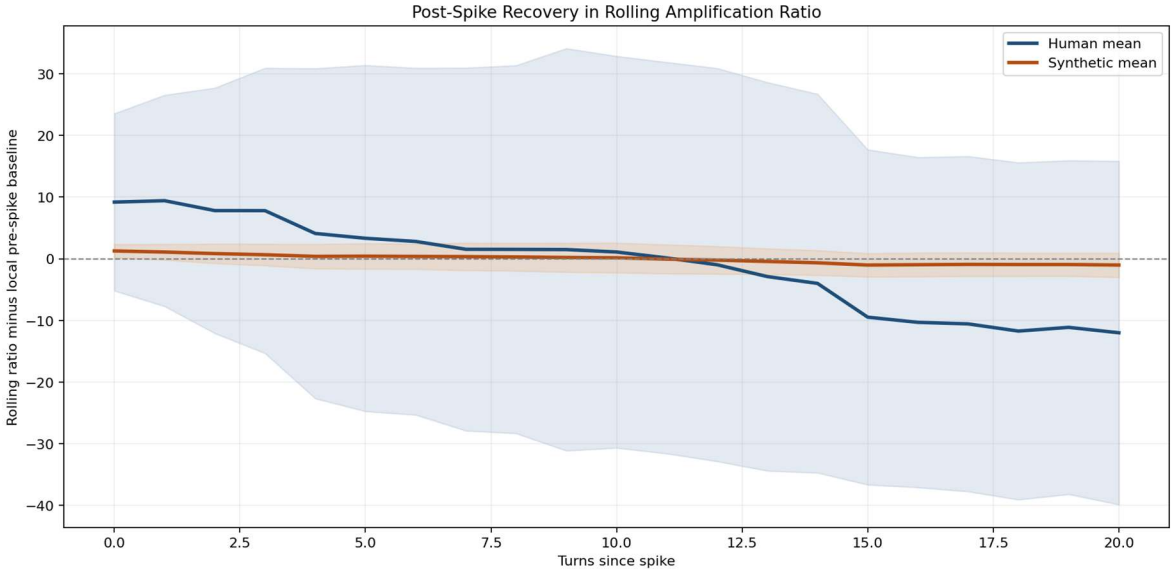

Figure 4. Post-spike recovery. Both conditions return toward range after spikes, but synthetic runs recover with less excursion and tighter regularization.

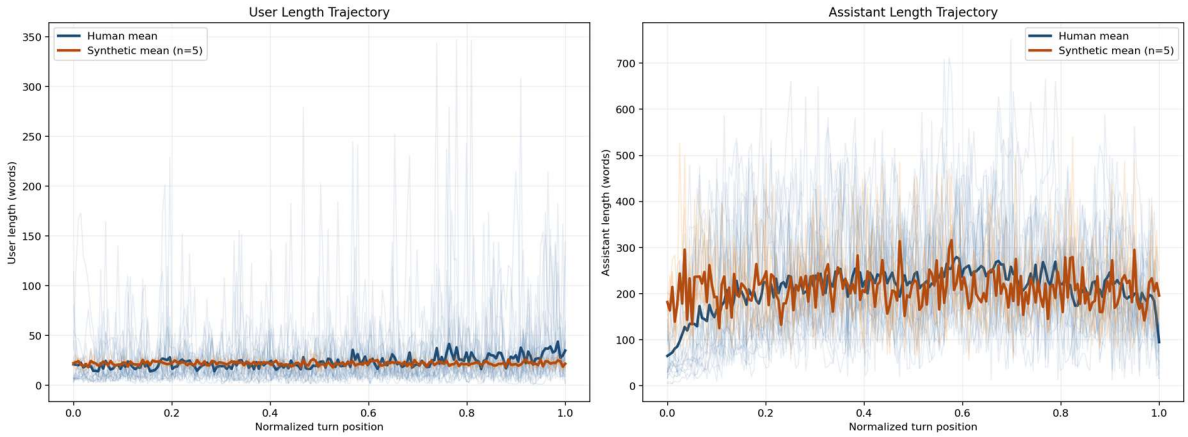

Figure 5. Normalized trajectory overlays. Mean synthetic paths remain flatter than the empirical archive despite close assistant-side bulk matching.

5. Methodological Finding

Constructing a strong null for long-form interaction required substantially more than matching first-order prompt statistics. Across repeated calibration attempts, synthetic runs could be brought close to the empirical assistant envelope while remaining systematically compressed on the user side and flatter on coupled metrics such as amplification.

This indicates that null construction in this setting is itself a nontrivial methodological problem, especially when the baseline is deliberately constrained to avoid user-specific imitation. The synthetic baseline was designed not to fail, but to isolate which properties of the empirical interaction are load-bearing for temporal morphology.

6. Discussion

The present comparison does not support the claim that broad surface matching alone is sufficient to reproduce the full temporal morphology of the empirical archive. Assistant-side bulk behavior can be approximated closely within a stateless synthetic baseline. User-side variability, amplification structure, and recovery dynamics are not recovered to the same degree.

We do not interpret this residual as evidence of semantics, model agency, or hidden learning. Within this archive and this baseline family, the result is narrower and more specific: substantial surface matching was insufficient. Closing the residual further would likely require injecting higher-order temporal dependence or user-specific adaptation, which would weaken the synthetic baseline's status as a matched control.

More broadly, the comparison suggests that no single matched statistic is sufficient to characterize the empirical regime. The observed structure depends on multiple coupled envelopes, including user length, assistant length, amplification behavior, local continuity, and long-horizon boundedness. As shown in Baldovino (2026), large-scale temporal structure is present in the archive. The present paper shows that a serious surface-matched stateless null does not fully recover that structure.

7. Limitations

The empirical archive is a single-user, fixed-boundary case study and should not be treated as a population claim. The analysis is structural rather than semantic. No universal claims are made about long-form interaction as a whole.

The synthetic runs were generated with the same API model family as the empirical archive, reducing model-family confounding. Remaining differences may still reflect interface, session, or condition-level factors in addition to differences in interaction structure.

The synthetic results reported here belong to one baseline family: a stateless, surface-matched generator with bounded local continuity. Stronger baselines could be constructed. Pushing the user-side synthetic materially closer risks turning the control into a disguised imitation of the focal archive.

Lagged coupling is treated as descriptive rather than causal, and the post-spike analysis should be interpreted as a bounded comparison of recovery dynamics rather than as evidence of memory or hysteresis.

8. Conclusion

The present study tested whether a surface-matched stateless synthetic baseline could reproduce the temporal morphology observed in a long-form Human-LLM archive. The baseline closely approximated assistant-side bulk behavior, but it did not recover the broader user-side variability, amplification structure, or recovery dynamics of the empirical archive. In this comparison, surface matching was therefore insufficient to reproduce the full temporal morphology of the archive**.**

9. Author Contributions & AI Assistance Disclosure

This manuscript presents original theoretical work and conceptual modeling by Kristina Baldovino. The author is solely responsible for the research design, analytical synthesis, and interpretive conclusions. Portions of the manuscript text, including grammar refinement and formatting, were prepared with assistance from an OpenAI and an Anthropic language model under the author's direct guidance. The models served only as a technical writing aid. All intellectual content and editorial decisions remain the sole responsibility of the author

Appendix A. Supplementary Figure

The lagged coupling plot is retained as supplementary evidence. It provides secondary, directionally supportive evidence for the manuscript's main argument.

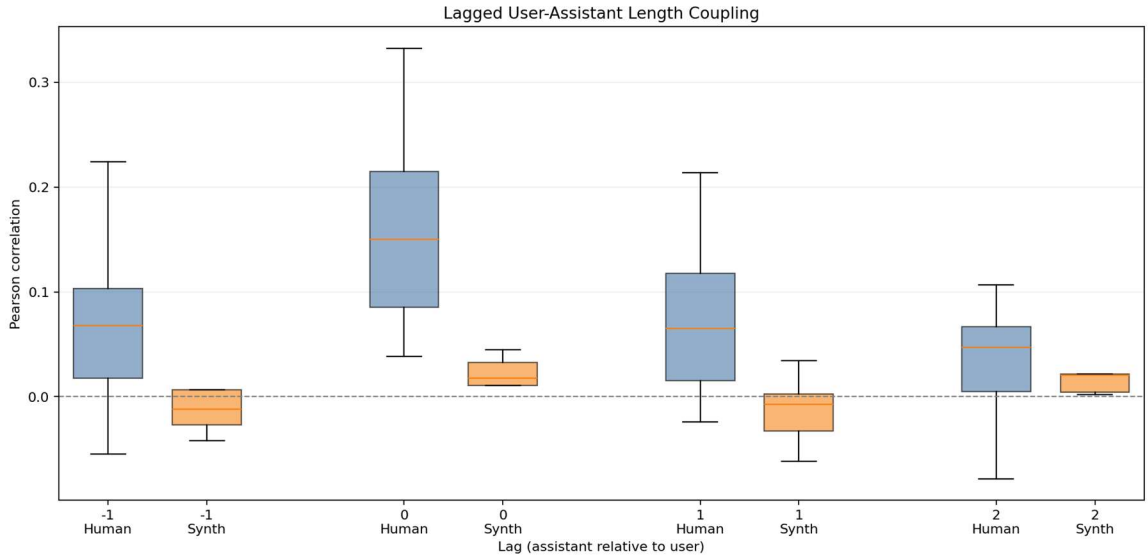

Figure A1. Lagged user-assistant length coupling. The empirical archive shows modestly stronger same-turn and short-lag association than the synthetic runs.

References

Baldovino, K. (2026). Beyond Content: Temporal Dynamics of Long-Form Human-LLM Interaction (v1.0). Zenodo. https://doi.org/10.5281/zenodo.18273459